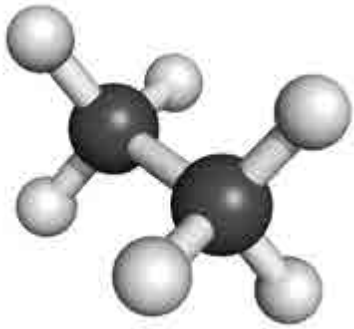
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Staggered conformation:



In this conformation, the hydrogens of both the carbon atoms are far apart from each other. The repulsion between the atoms is minimum and it is the most stable conformer.

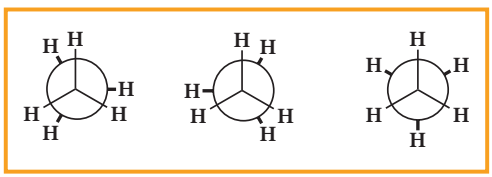
Skew Conformation :

The infinite numbers of possible intermediate conformations between the two extreme conformations are referred as skew conformations.

The stabilities of various conformations of ethane are

Staggered > Skew > Eclipsed

The potential energy difference between the staggered and eclipsed conformation of ethane is around 12.5 $KJmol^{-1}$. The various conformations can be represented by new man projection formula.



Eclipsed Skew Staggered

Newman projection formula for Ethane

Conformations of n-Butane:

n-Butane may be considered as a derivative of ethane, as one hydrogen on each carbon is replaced by a methyl group

Eclipsed conformation:

In this conformation, the distance between the two methyl group is minimum. So there is maximum repulsion between them and it is the least stable conformer.

Anti or staggered form

In this conformation, the distance between the two methyl groups is maximum and so there is minimum repulsion between them. And it is the most stable conformer.

The following potentially energy diagram shows the relative stabilities of various conformers of n-butane.

